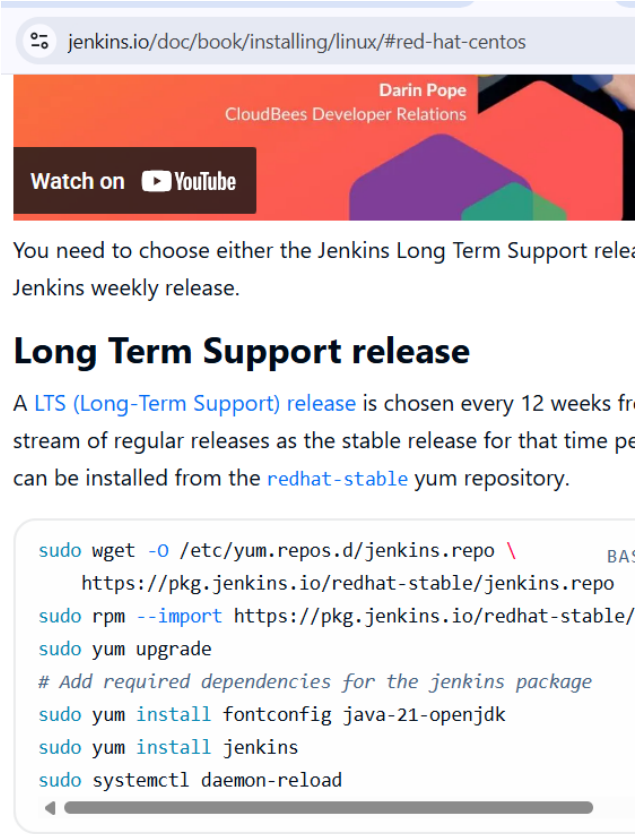
JENKINS 01

1 . Install Jenkins and run Jenkins on port number 8081.

- Open aws console
- Launch an ec2 instance
- Install Jenkins
- > browser search for Jenkins install for linux environment



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- Copy the command and paste in ec2 instance Jenkins will be installed

```
  Version 2023.9.20251105:
    Run the following command to upgrade to 2023.9.20251105:

      dnf upgrade --releasever=2023.9.20251105

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.
.20251105.html

  Version 2023.9.20251110:
    Run the following command to upgrade to 2023.9.20251110:

      dnf upgrade --releasever=2023.9.20251110

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.
.20251110.html

================================================================================

Installed:
  jenkins-2.528.1-1.1.noarch

Complete!
```

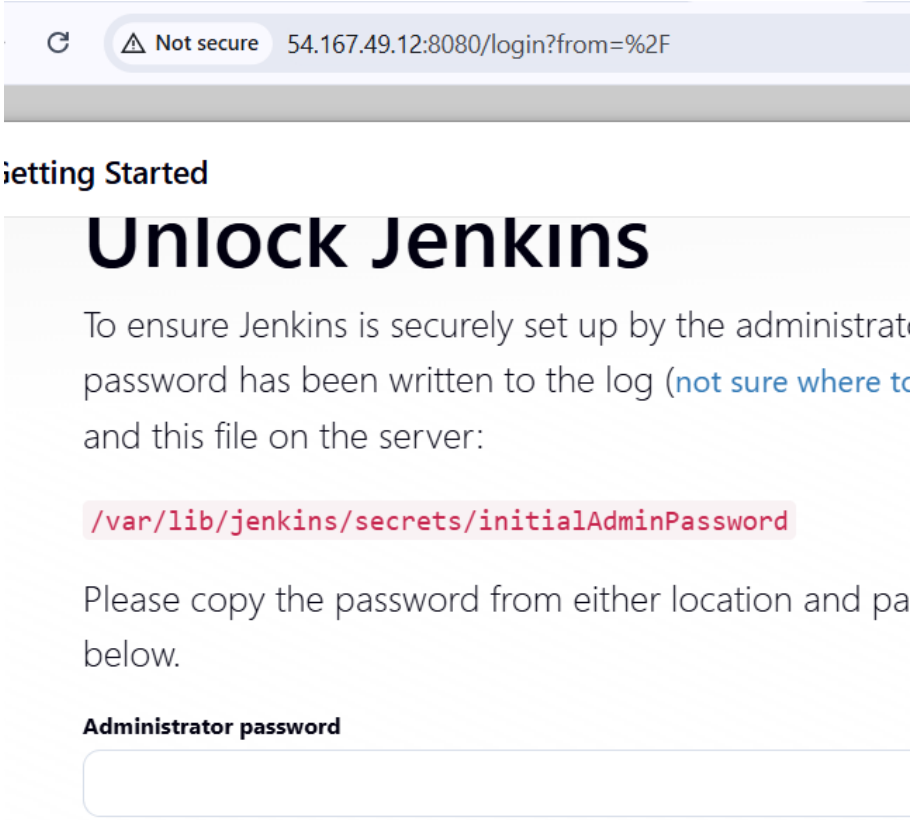
- sudo wget -O /etc/yum.repos.d/jenkins.repo \
- https://pkg.jenkins.io/redhat-stable/jenkins.repo
-
- sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
- sudo yum upgrade
- # Add required dependencies for the jenkins package
- Install java
- sudo dnf install java-17-amazon-corretto -y

```
Complete!
[root@ip-10-0-1-110 ec2-user]# java -version
openjdk version "17.0.17" 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixe
 mode, sharing)
[root@ip-10-0-1-110 ec2-user]#
```

- sudo yum install jenkins
- systemctl start Jenkins

```
[root@ip-10-0-1-110 ec2-user]# systemctl status jenkins.service
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; pr
     Active: active (running) since Tue 2025-11-11 09:19:49 UTC; 4s ago
   Main PID: 3555 (java)
      Tasks: 40 (limit: 1053)
     Memory: 359.1M
        CPU: 20.439s
     CGroup: /system.slice/jenkins.service
             └─3555 /usr/bin/java -Djava.awt.headless=true -jar /usr/share

Nov 11 09:19:45 ip-10-0-1-110.ec2.internal jenkins[3555]: [LF]> This may a
Nov 11 09:19:45 ip-10-0-1-110.ec2.internal jenkins[3555]: [LF]>
Nov 11 09:19:45 ip-10-0-1-110.ec2.internal jenkins[3555]: [LF]> **********
Nov 11 09:19:45 ip-10-0-1-110.ec2.internal jenkins[3555]: [LF]> **********
Nov 11 09:19:45 ip-10-0-1-110.ec2.internal jenkins[3555]: [LF]> **********
Nov 11 09:19:49 ip-10-0-1-110.ec2.internal jenkins[3555]: 2025-11-11 09:19
Nov 11 09:19:49 ip-10-0-1-110.ec2.internal jenkins[3555]: 2025-11-11 09:19
Nov 11 09:19:49 ip-10-0-1-110.ec2.internal systemd[1]: Started jenkins.ser
Nov 11 09:19:50 ip-10-0-1-110.ec2.internal jenkins[3555]: 2025-11-11 09:19
Nov 11 09:19:50 ip-10-0-1-110.ec2.internal jenkins[3555]: 2025-11-11 09:19
lines 1-20/20 (END)
```

- 

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- Now, navigate to default location of Jenkins = /var/lib/Jenkins
- Find secrets
- Go to secrets location and list them
- And cat the initinal password

```
[root@ip-10-0-1-110 ec2-user]# ^C
[root@ip-10-0-1-110 ec2-user]# cd /var/lib/jenkins
[root@ip-10-0-1-110 jenkins]# ls
config.xml                        nodeMonitors.xml          secrets
hudson.model.UpdateCenter.xml     plugins                   updates
jenkins.telemetry.Correlator.xml  secret.key                userContent
jobs                              secret.key.not-so-secret  users
[root@ip-10-0-1-110 jenkins]# cat secret
cat: secret: No such file or directory
[root@ip-10-0-1-110 jenkins]# cd secrets/
[root@ip-10-0-1-110 secrets]# ls
hudson.model.User.DIRNAMES  jenkins.model.Jenkins.crumbSalt
initialAdminPassword        master.key
[root@ip-10-0-1-110 secrets]# cat initialAdminPassword
a3c4d613227b4db08466902a3aebacca
[root@ip-10-0-1-110 secrets]#
```

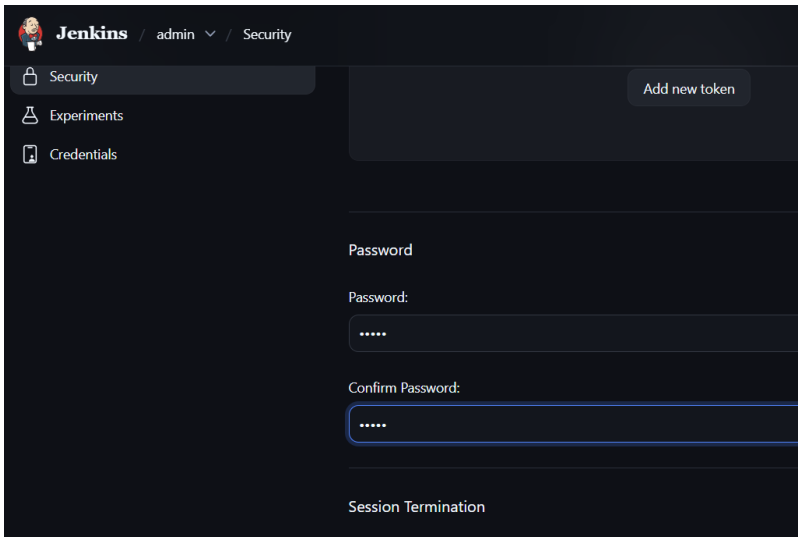
- Now the copy the password and paste it in Jenkins page ,
- Now to change port number
- Navigate to this location.
- sudo vi /usr/lib/systemd/system/jenkins.service
- search for environment and change port number.
-

```
# IP address to listen on for HTTP requests.
# The default is to listen on all interfaces (0.0.0.0).
#Environment="JENKINS_LISTEN_ADDRESS="

# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"

# IP address to listen on for HTTPS requests. Default is disabled.
#Environment="JENKINS_HTTPS_LISTEN_ADDRESS="

# Port to listen on for HTTPS requests. Default is disabled.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
#Environment="JENKINS_HTTPS_PORT=443"

# Path to the keystore in JKS format (as created by the JDK's keytool).
# Default is disabled.
#Environment="JENKINS_HTTPS_KEYSTORE=/path/to/keystore.jks"
```

- Now restart Jenkins
- Systemctl daemon-reload
- Now,
- Systemctl restart Jenkins
- Now.,Jenkins running on port number 8081
- Allow inbound rules form 8081 port in ec2

2 . Secure Jenkins server

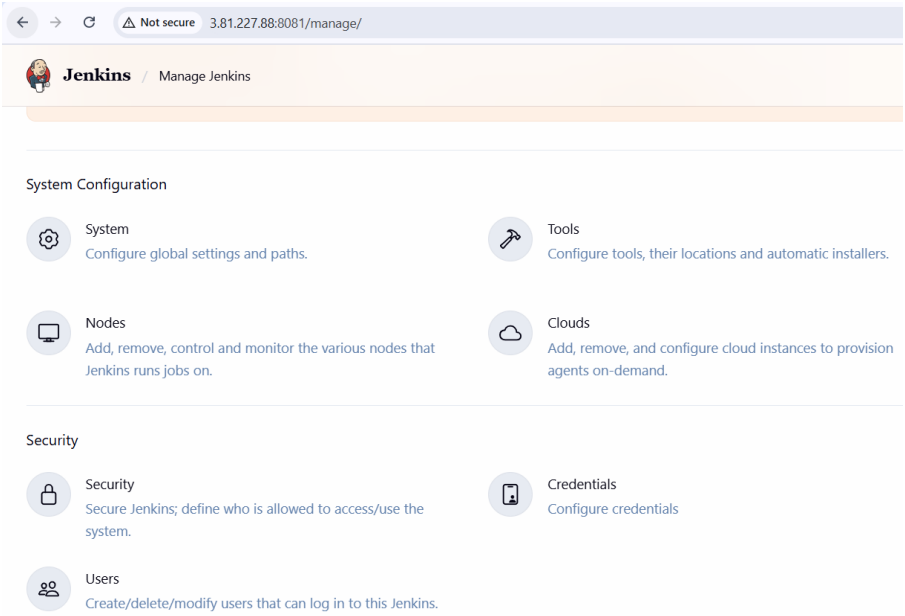
- Copy password from
  sudo cat /var/lib/jenkins/secrets/initialAdminPassword



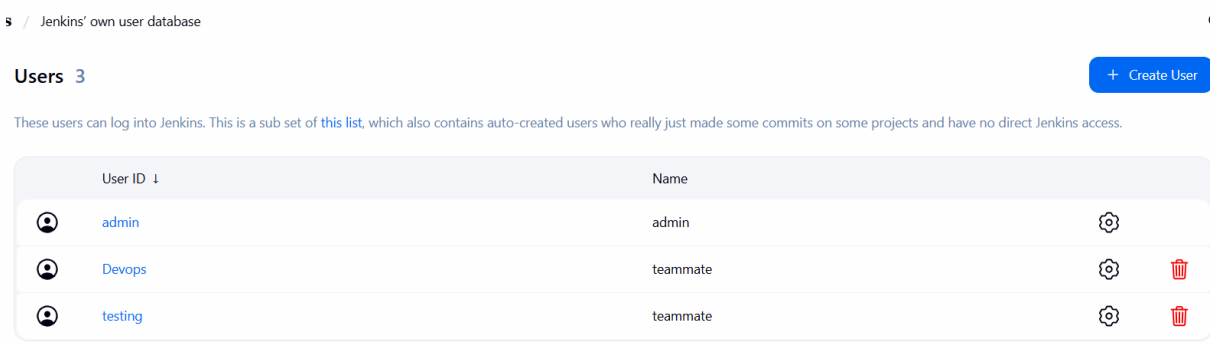
- Add new password and user name and save it

3 . Create users called DevOps, Testing in Jenkins with Limited access.

- Login to Jenkins
- Navigate to manage Jenkins



- Select user for creating user
- Add user name , password and member name



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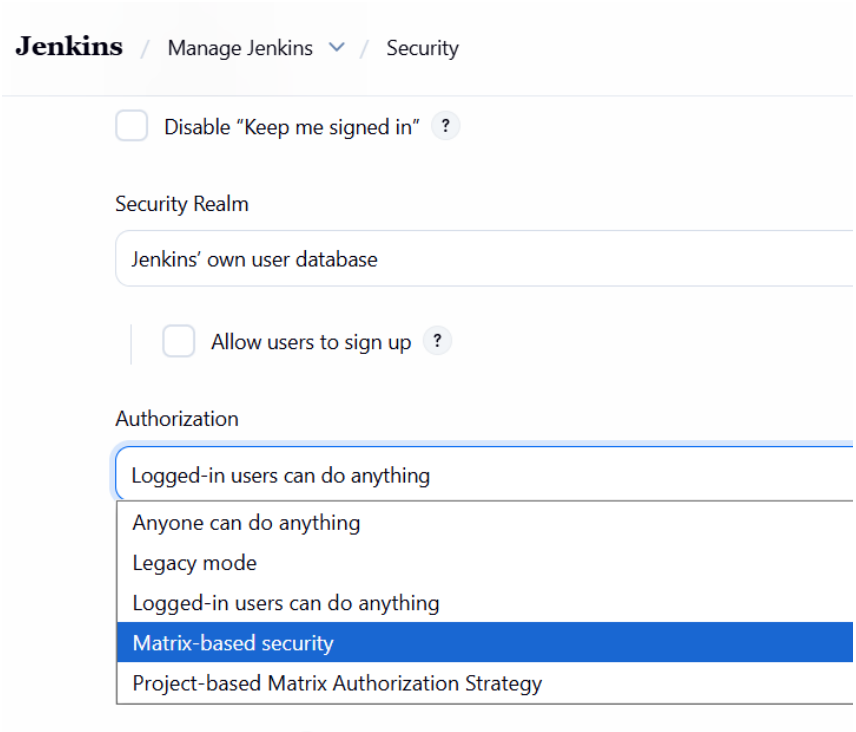
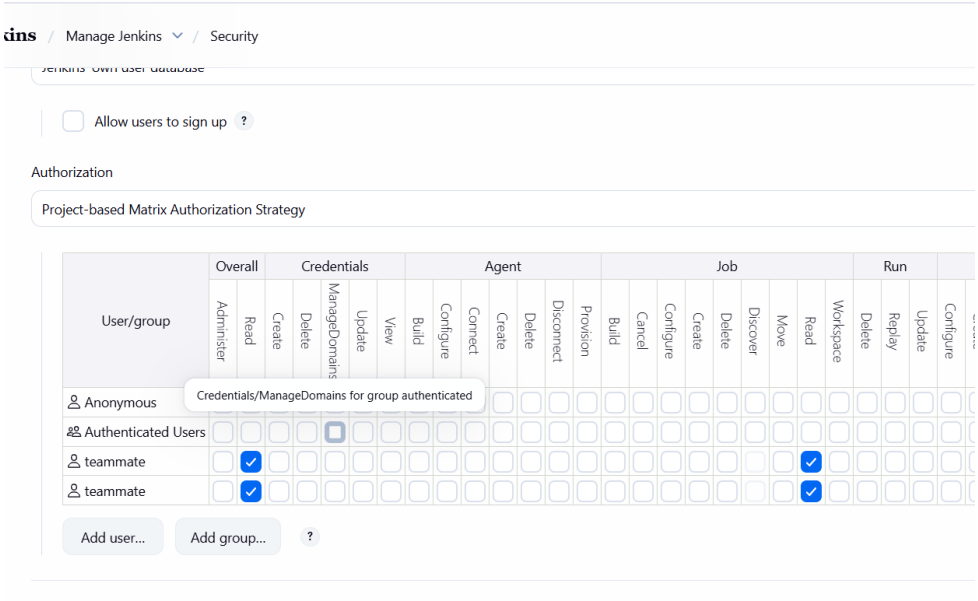
- User created.
- Now giving limited acces to devops and tester
- Navigate to manage Jenkins > and the n security and find for authorization and select > project-based matrix authorization strategy

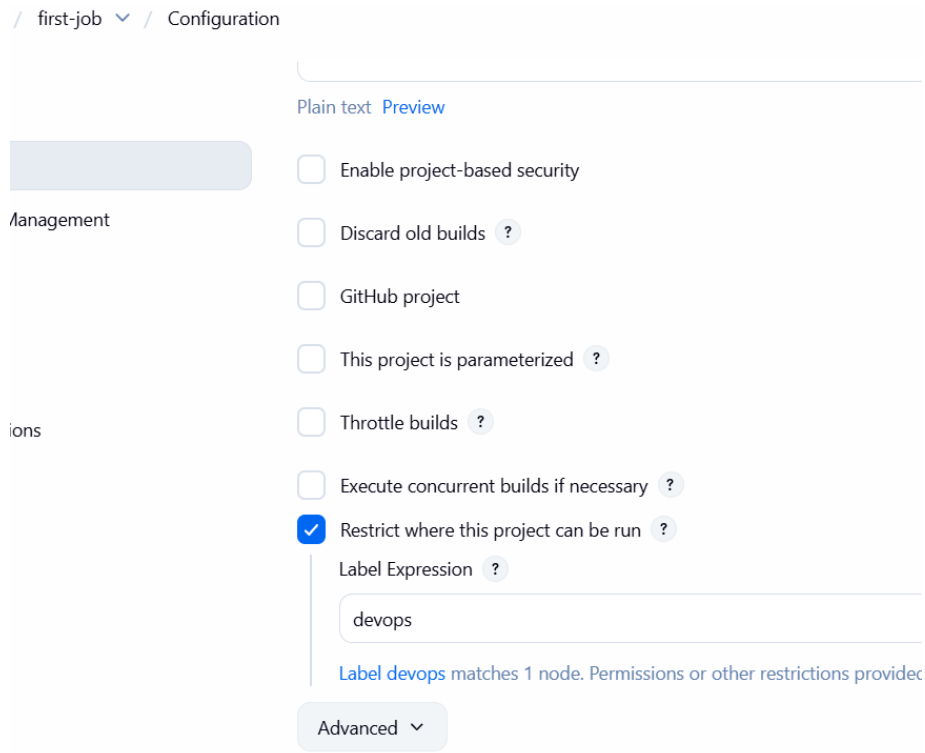




-
- Gave read acces to devops and tresting users

4 . Configure labels and restrict the jobs to execute based on label only.

- Navigate to manage Jenkins
- Open nodes
- Add label name as devops
- Now downloads pulgins for node js
- And go free style project and restrict where this project can be run



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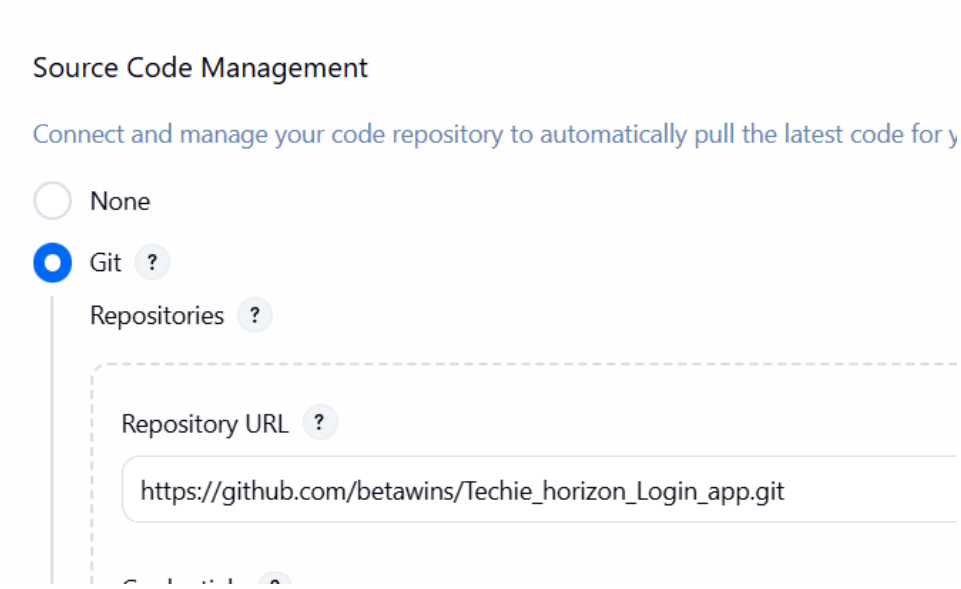
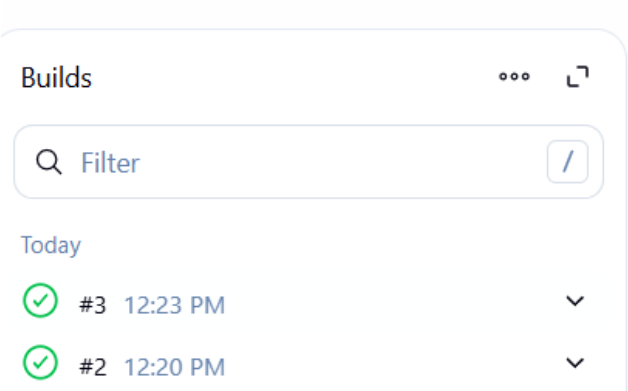
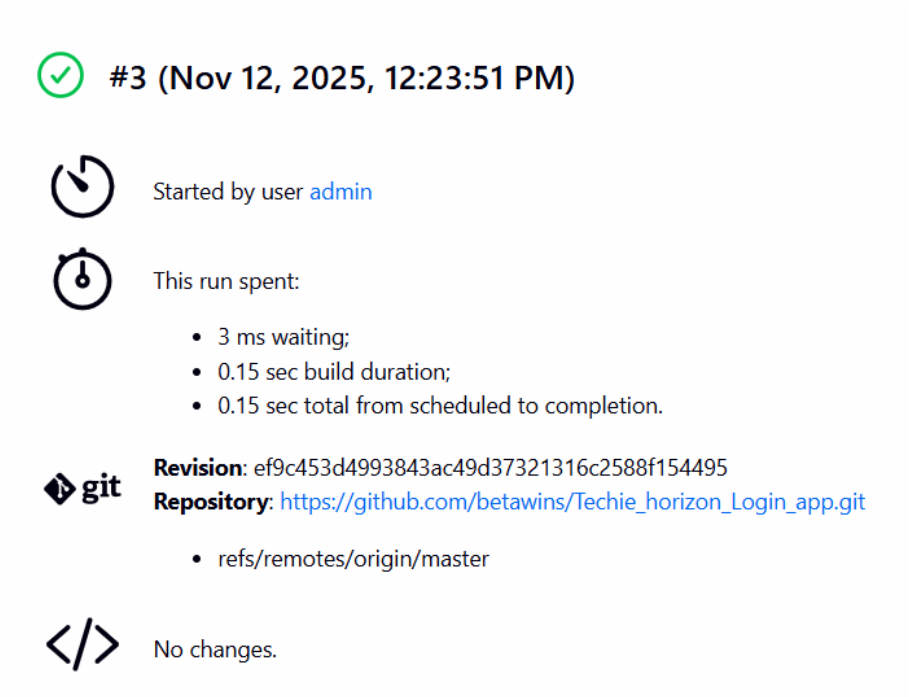
5 . Create Three sample jobs using the below URL. https://github.com/betawins/Techie_horizon_Login_app.git

- Add git in to Jenkins
- Install git in terminal
- Now, give git path in tools
- /user/bin/git
- Copy the git repo
- Add to source code management



- Sechudle a build
- Start building now

- 

- 

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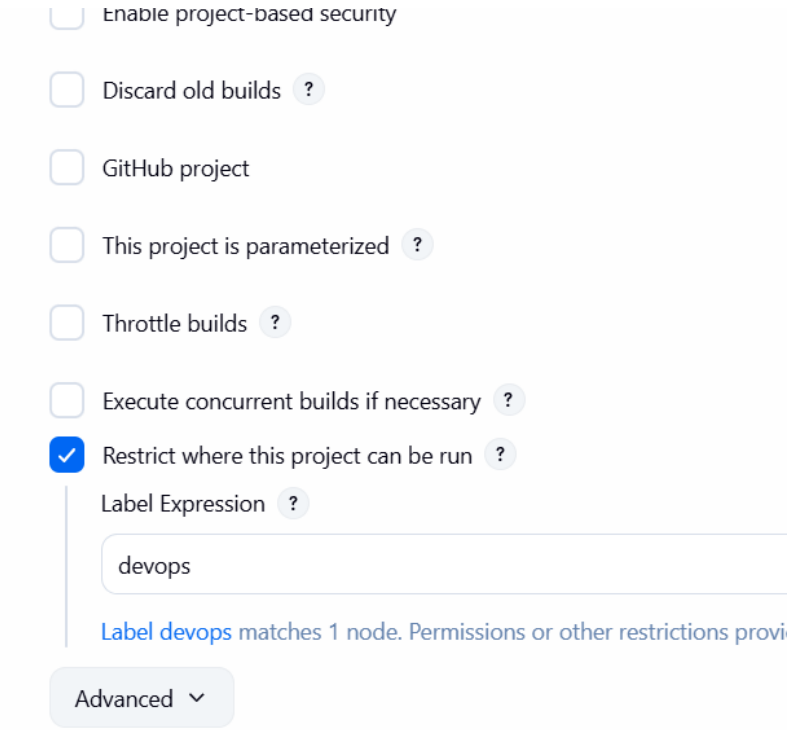
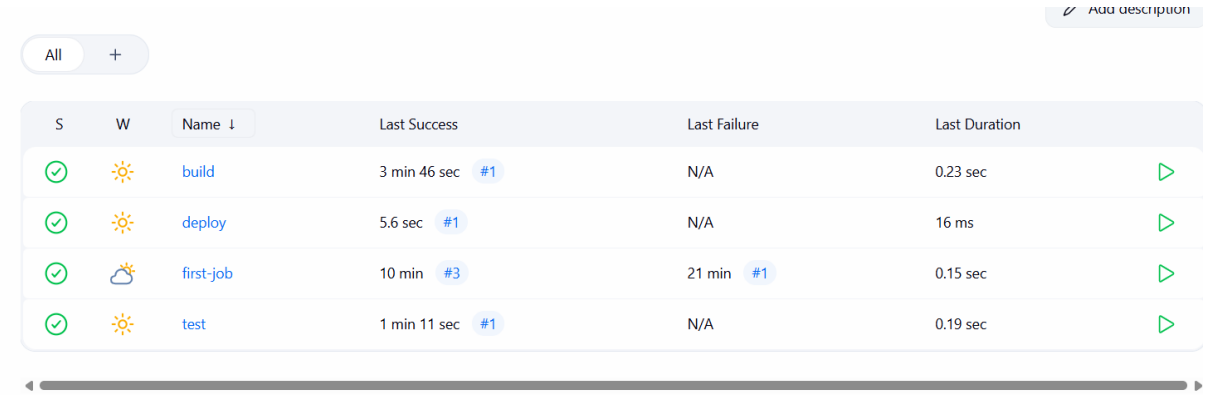
- Create 3 jobs
- Navigate to Jenkins
- Create a new item
- Add name
- Select the - restrict where this project can be run



-
- Add git repo
- Add branch
- Add time stamps, terminate if it stucks



6 . Create one Jenkins job using git hub Private repository.

- Connect Jenkins and github
- By generating ssh keygen
- Switch to Jenkins user
- sudo su – Jenkins
- now. Generate key
- ssh-keygen -t rsa -b 4096 -C "jenkins"
- save the key
- to view the key

- cat ~/.ssh/id_rsa.pub

```
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:7OeVLBcV1QT0uEKQI3dYZDaliR228x+YW8LBW3VnBMc jenkins
The key's randomart image is:
+---[RSA 4096]----+
|          .=Oo*B@|
|        . =Bo* *E|
|         o.oBoo..|
|       .   ..+*. |
|        S   o*oo |
|       .   . ++..|
|        . o =.  .|
|         o +     |
|          .      |
+----[SHA256]-----+
[ec2-user@ip-10-0-1-110 ~]$ cat ~/.ssh/id_rsa.pub
```

- now copy key and past e in github deploy
- now Jenkins ,
- go to manage Jenkins
- credentials add git repo
- and add secret key to it

Credentials

T	P	Store ↓	Domain	ID	Name
		System	(global)	58670c12-0222-4f46-87d9-fda93cf8640f	git

- now create a job add git repo
- add ssh git in the section
- apply and save it
- start building

Console Output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/jenkins
The recommended git tool is: NONE
Warning: CredentialId "github" could not be found.
 > /usr/bin/git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/jenkins/.git # timeout
Fetching changes from the remote Git repository
 > /usr/bin/git config remote.origin.url git@github.com:kamal-kkk/local-to-github.git # timeo
Fetching upstream changes from git@github.com:kamal-kkk/local-to-github.git
 > /usr/bin/git --version # timeout=10
 > git --version # 'git version 2.50.1'
 > /usr/bin/git fetch --tags --force --progress -- git@github.com:kamal-kkk/local-to-github.g
 > /usr/bin/git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision b1aeaf26c71ef741cc806cd3f3441ba43cdd3d16 (refs/remotes/origin/master)
 > /usr/bin/git config core.sparsecheckout # timeout=10
 > /usr/bin/git checkout -f b1aeaf26c71ef741cc806cd3f3441ba43cdd3d16 # timeout=10
Commit message: "your update"
First time build. Skipping changelog.
Finished: SUCCESS
```